

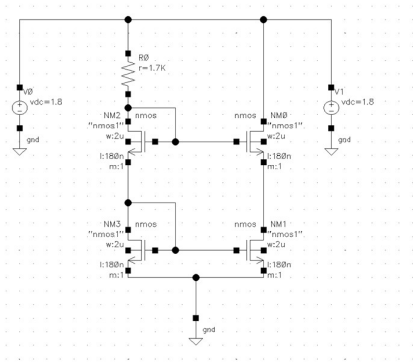
Name : Rahul Bhadane

Roll no: 23104B0003

EXPERIMENT 3

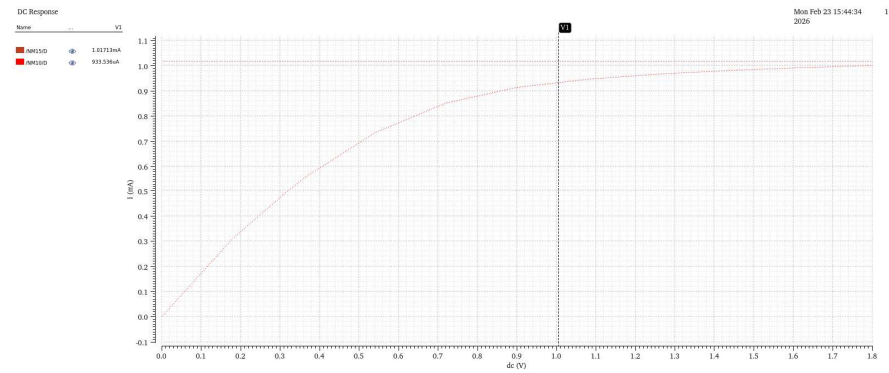
Aim: Design and simulation of Cascode current source using 180nm NMOS TX.

Circuit Diagram:

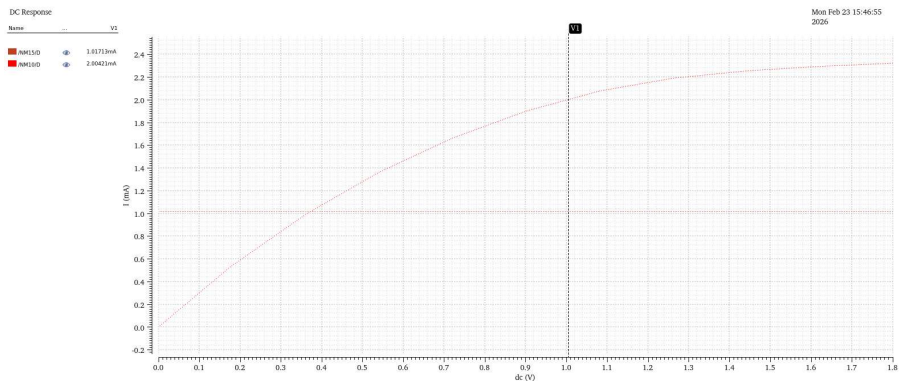


Results:

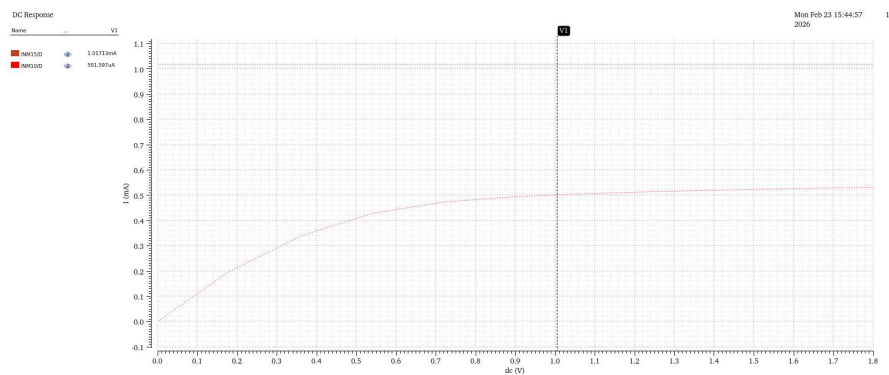
1mA using 1mA Iref



2mA using 1mA Iref



0.5mA using 1mA Iref



Conclusion: The simulation of the NMOS cascode current source in 180 nm technology verified its ability to deliver a more constant current with higher output resistance. The additional cascode transistor minimized variations due to voltage changes, thereby improving current regulation and overall performance in precision analog designs.